Muhammed Irshad Ismail

Software Engineering / Site Reliability Engineering Intern Candidate | Python, Backend, Distributed Systems
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Expected graduation: 2028 (Penultimate year, B.Sc.) | Available full-time for 13+ weeks, Summer 2027 (May-July)

PROFESSIONAL SUMMARY

Computer science-focused Artificial Intelligence student with 2+ years of hands-on experience building production backend systems and APIs in Python. Solid foundation in data structures and algorithms (250+ LeetCode problems), distributed systems, and Linux-based infrastructure (Docker, Kubernetes, CI/CD, AWS). Experienced in message-driven architectures, asynchronous processing, automation, and testing, with open-source contributions to production Python projects. Seeking the Google Software Engineering / SRE Intern role in Warsaw to build reliable, scalable systems.

EDUCATION

Bachelor's Degree in Artificial Intelligence, Vistula University, Warsaw 2025 - Present (Expected 2028)

3.5-year full-time program | Penultimate year | Relevant coursework: Advanced Python & Web Development, SQL & RDBMS, Services Architecture & Patterns, Testing, Message Brokers, NoSQL, DevOps, Cloud Platforms | Languages: English (Fluent), Malayalam (Native)

TECHNICAL SKILLS

Languages: Python 3, SQL, JavaScript, TypeScript, Bash/Shell
CS Fundamentals: Data Structures & Algorithms (graphs, dynamic programming, greedy, trees), System Design, Distributed Systems, Asynchronous Programming
Infrastructure & SRE: Linux/Unix, Docker, Kubernetes, CI/CD (GitHub Actions, GitLab), AWS, Automation Scripting, Failure Handling, Reliability & Scalability
Backend & Data: FastAPI, Django REST Framework, Django Ninja, Flask, REST API Design, Celery, WebSockets, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Query Optimization
AI / ML: LLM API Integration, AI Agent Frameworks, Prompt-Driven Workflows, Structured Output Validation, Conversational AI
Engineering Practices: pytest, Unit/Integration/API Testing, TDD, Clean Architecture, Code Review, Git, Agile/Scrum
Exposure: Kafka, Airflow, GCP (learning)

PROFESSIONAL EXPERIENCE

Python Engineer Intern, Innowise | Poland Jul 2026 - Present

- Building a distributed microservices platform in Python (User Management, Notification, and Innotter services) using REST APIs, Clean Architecture, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Docker, and Kubernetes.
- Implementing message-driven workflows with RabbitMQ and Redis to improve reliability, scalability, and data consistency, including failure handling across services.
- Designing service boundaries, data models, and synchronous/asynchronous communication flows; integrating AWS services in a multi-service environment.
- Writing unit, API, and integration tests with pytest; collaborating via Git pull requests, code review, and Agile workflows.

Full-Stack Software Engineer (Freelance), Code Cave | Warsaw, Poland Jun 2023 - Feb 2026

- Designed, built, and shipped production REST APIs in Python (FastAPI, Django REST Framework) on PostgreSQL for client and merchant workloads.
- Architected asynchronous job pipelines with Celery and Redis, moving long-running work off synchronous API paths to improve reliability and scalability.
- Optimized PostgreSQL schemas and SQL queries for transactional workflows; contributed backend performance optimizations in a large production Python codebase.
- Containerized development and deployment environments with Docker; collaborated through Git/GitLab feature branching, pull requests, and code reviews.

SELECTED PROJECTS

Distributed Microservices Platform (In Progress)

Stack: Python 3, FastAPI, PostgreSQL, MySQL, MongoDB, Redis, RabbitMQ, Docker, Kubernetes, AWS S3, AWS SES

- Designed a service-oriented platform connecting User Management, Innotter, and Notification services via synchronous HTTP and asynchronous message-broker flows; covers authentication, caching, relational and NoSQL persistence, and object storage.

Mindorax - AI-Powered Conversational Assistant

Stack: Python, Django Ninja, PostgreSQL, LLM API Integration, TypeScript, Docker

- Built a Python backend for an AI conversational application that processes user queries and returns structured AI responses with scope validation.
- Applied clean architecture and modular service design, and added unit tests and validation for edge cases to improve correctness and reduce regressions.

OPEN SOURCE & PROBLEM SOLVING

- Active contributor to production Python open-source projects, including an iCalendar library and an AI agent orchestration framework; submitted code improvements, resolved issues, and collaborated through GitHub/GitLab pull requests.
- Solved 250+ LeetCode problems covering graphs, dynamic programming, greedy algorithms, and trees.